Bose Beamline Challenge (BBLC) 2026

Research Proposal | Simulation-Backed Beamline Study

σ scanner

Turning Proton Scattering into a Second Imaging Signal for Safer Range Decisions in Proton Therapy

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Instrumentation
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Method

Geant4 11.4.0 simulation, Highland scattering model, Monte Carlo statistics validation, NIST PSTAR stopping powers, and bootstrap uncertainty analysis.

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Abstract

Proton therapy works because a proton beam stops sharply at a chosen depth, and planning that depth safely depends on knowing the material the beam passes through. Two tissues can lose almost exactly the same energy to the beam yet be made of different elements, and ordinary proton imaging, which reads energy loss, cannot tell them apart. We propose to image instead with a signal that most analyses throw away: the small, repeated deflections a proton picks up as it scatters. Because scattering depends on atomic composition rather than on energy loss, it can separate materials that energy loss leaves identical. We test this on two plastics, PVC and PEEK, that lose energy almost identically but differ in composition because PVC contains chlorine. Using a proton beam at the Bose Beamline Challenge facility, a telescope of tracking chambers, and a rotating plastic phantom, we would measure how widely each proton scatters. A detailed Geant4 simulation, prepared here as supporting evidence, shows the two plastics staying indistinguishable in energy loss while separating cleanly and unambiguously in scattering, in close agreement with established theory. A beam test would confirm that this contrast holds under real conditions and establish scattering as a second, complementary way to see inside matter, for safer range decisions in proton therapy.

1) Introduction

Proton therapy is a powerful radiotherapy modality because charged particles deposit most of their dose in a localized volume and terminate with a sharp Bragg peak, helping to spare nearby organs at risk. That same advantage makes accurate knowledge of the material budget along the beam path essential. If tissue properties are misestimated, the planned proton range can shift by **amounts large enough to affect treatment** [1–3].

Conventional X-ray computed tomography provides an indirect input to proton therapy planning (Figure 1). X-ray attenuation depends on electron density, whereas proton transport depends on stopping power and range. Proton computed tomography (pCT) addresses that mismatch more directly by measuring proton transmission and reducing the conversion bias [4, 5].

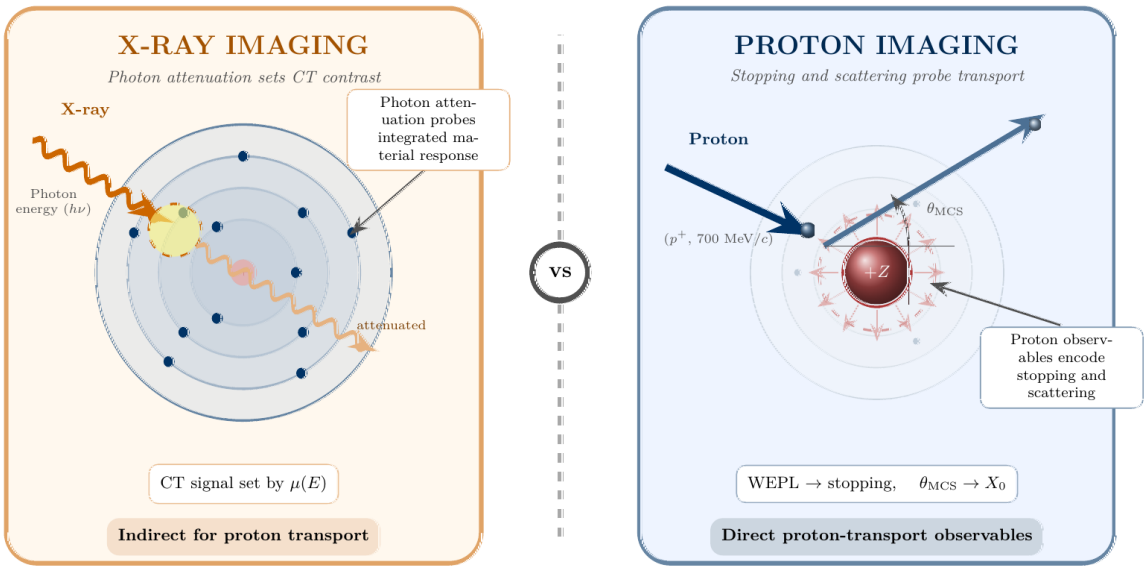


Figure 1. *Schematic comparison of imaging channels. Conventional X-ray CT measures photon attenuation and must be converted into proton-transport quantities for treatment planning. Proton imaging instead accesses stopping and scattering observables along the beam trajectory. The figure is qualitative rather than a quantitative scaling comparison.*

However, multiple Coulomb scattering (MCS) is usually treated as a noise rather than as the signal. Repeated deflections blur the proton path and limit spatial precision, so reconstruction pipelines typically try to suppress scattering in order to recover accurate stopping information. Scattering-power CT has also been shown to carry substantially higher noise than energy-loss imaging [6]. Yet those same fluctuations contain material-specific information: their statistics are governed by radiation length and therefore by composition in a way that is complementary to stopping power [7–9].

This study inverts the usual perspective by **treating scattering fluctuations themselves as the measurement signal**. Rather than relying on mean energy loss alone, we characterize the scattering-angle distribution and test whether materials that are difficult to separate by stopping-power-based imaging can instead be distinguished through their scattering response. X-ray attenuation and water-equivalent path length (WEPL) provide limited contrast for such pairs because neither observable isolates radiation-length differences directly [10, 11].

Therapeutic-energy proton scattering has been measured in other materials [12], but **no one has measured the PVC–PEEK pair in detail** in this energy regime.

This motivates our choice of polyvinyl-chloride (PVC) and polyether-ether-ketone (PEEK) as a test case. PVC and PEEK form a challenging pair to tell apart: they are **nearly degenerate in stopping-power-based imaging but separate in the scattering channel** because PVC contains chlorine. Chlorine has a higher atomic number (Z = 17) than the carbon, hydrogen, and oxygen of PEEK, which shortens the radiation length and makes the beam scatter more. The aim is to test whether a realistic proton-imaging measurement can recover that scattering contrast under beamline conditions.

2) Objectives and Hypothesis

Research questions.

1. Can two materials that are nearly degenerate in stopping power (PVC and PEEK) be separated using only the width of their multiple-Coulomb-scattering distribution?
2. Does the measured PVC/PEEK scattering-width ratio agree with the Highland prediction based on radiation length?
3. *(Stretch)* Can the scattering signal be reconstructed into a two-dimensional material map of the phantom?

Hypothesis. Because PVC contains chlorine (Z = 17) it has a shorter radiation length than PEEK, so a proton beam will scatter measurably more in PVC even though the two materials lose almost the same energy. Quantitatively, we predict a scattering-width ratio of about 1.34 (Highland) and therefore a clear, statistically significant separation between the two materials.

Null hypothesis. PVC and PEEK produce no statistically significant difference in scattering-angle width once the common beamline and air background is subtracted.

Objectives. Our *primary* objective is to demonstrate PVC–PEEK discrimination from the aggregate full-phantom run at a single momentum. Our *secondary* objectives are to validate the Highland scaling against a momentum scan and to test for systematic biases through an insert-reversal run; a tomographic reconstruction of the phantom is a stretch goal.

3) Physics Background

Proton transmission carries two complementary observables: energy loss, which constrains WEPL and stopping power [10, 11], and **the scattering-angle width, which probes radiation length** [8, 9]. Both observables are routinely combined in proton-CT pipelines [3, 5].

As protons traverse matter, repeated Coulomb deflections in the fields of atomic nuclei generate a narrow central angular distribution whose width encodes the material response (Figure 2) [7–9]. The key material property here is the radiation length X_0: the smaller X_0, the more the beam spreads.

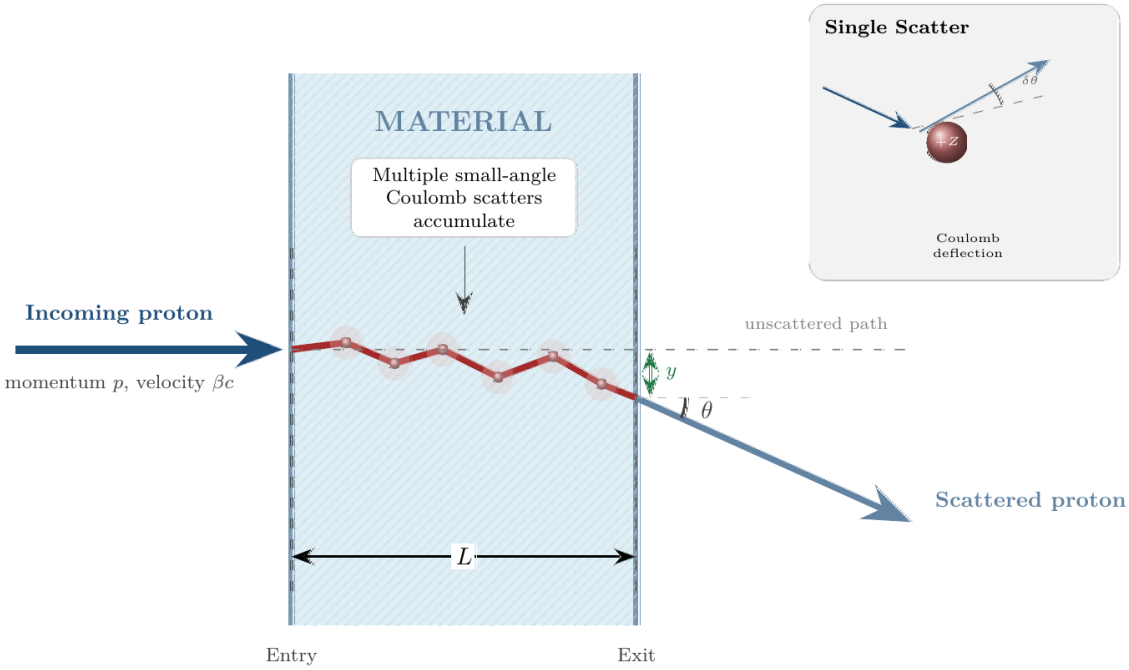


Figure 2. *Geometry of multiple Coulomb scattering through a slab of thickness L. Many small Coulomb deflections broaden the exit direction by an angle θ relative to the unscattered path. The characteristic width of the central angular distribution scales with √(L/X₀), linking the observable directly to radiation length.*

The Highland-Lynch-Dahl formula [13, 14] quantifies this relationship:

$$\theta_0 = \frac{13.6\,\mathrm{MeV}}{\beta\, p\, c}\sqrt{\frac{x}{X_0}}\left[1 + 0.038\,\ln\left(\frac{x}{X_0}\right)\right]$$

where $\beta = v/c$ is the proton velocity, p the momentum, x the material thickness, and X_0 the radiation length. For a fixed beam and thickness, the dominant scaling is approximately $\theta_0 \propto \sqrt{(x/X_0)}$, so **materials with smaller radiation length produce broader scattering**, the central first-order behaviour that grounds this study's PVC/PEEK comparison [8, 9]. Rare nuclear interactions populate the large-angle tails of the scattering distribution. For that reason, the material comparison in this study is based on the Coulomb-dominated central width, which is the part most directly linked to radiation length [9].

The repeat-unit chemistry of the two primary polymers makes the radiation-length difference explicit. As shown in Figure 3, **PVC introduces chlorine (Z = 17) into the polymer, lowering X_0 to 18.2 cm**, whereas PEEK is built from carbon, hydrogen, and oxygen only (Z_{max} = 8), giving X_0 = 31.4 cm. This chemistry changes radiation length much more than stopping power, which is why PVC and PEEK form a meaningful scattering test pair [8–10]. For a 5 cm insert at the 0.7 GeV/c working point of Section 4, the Highland formula predicts that PVC should scatter about 1.34 times as strongly as PEEK, the benchmark against which the simulated ratio in Section 5 is compared.

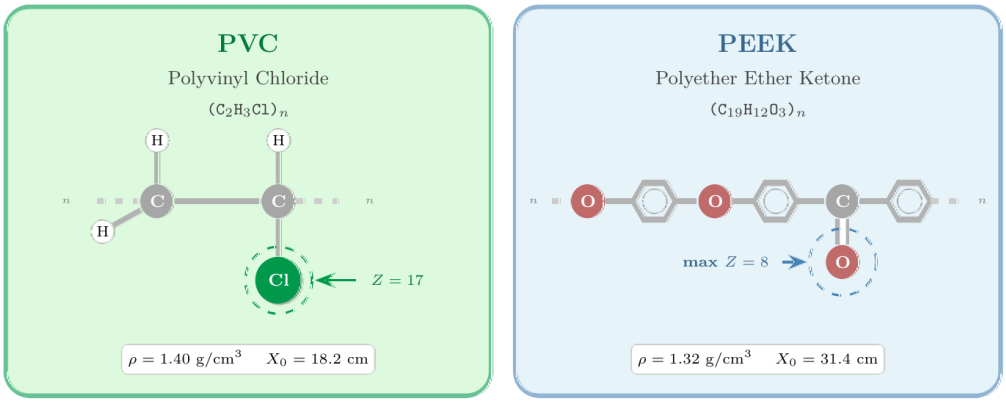


Figure 3. *Repeat-unit structures of the two primary polymer inserts. PVC contains chlorine, while PEEK is built from carbon, hydrogen, and oxygen only. Despite nearly identical WEPL values, this difference in elemental composition shifts the radiation length and motivates the PVC–PEEK comparison.*

The PVC–PEEK pair shows this directly: **WEPL is nearly degenerate, while scattering separates the materials** (Figure 4).

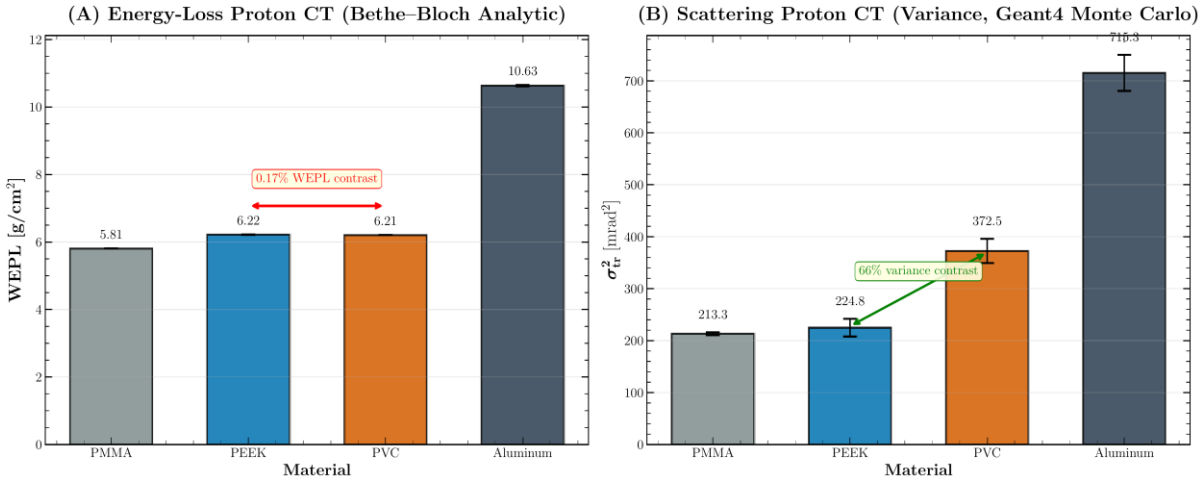


Figure 4. *Comparison of stopping and scattering observables for the primary polymer pair. (a) WEPL values from Bethe–Bloch: PVC and PEEK are nearly degenerate. (b) Geant4 scattering-variance contrast: the plotted (raw) PVC/PEEK variance ratio is about 1.66; after empty-target subtraction the differential ratio reported in Section 5 is 1.683.*

4 Experimental Design

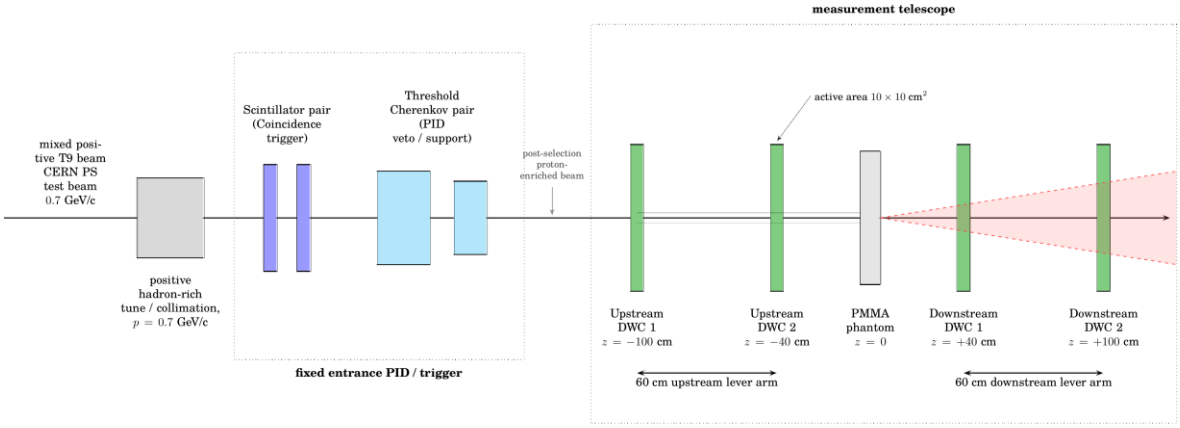


Figure 5. *Schematic of the proposed BBLC beamline setup. A mixed positive secondary beam is tuned to p = 0.7 GeV/c and filtered by the fixed scintillator trigger and threshold Cherenkov PID chain before entering the two-upstream/two-downstream DWC telescope around the PMMA phantom. The Geant4 model starts from this post-selection proton-enriched sample and treats the common upstream chain as passive material.*

The proposed measurement uses a **proton-enriched 0.7 GeV/c sample at the BBLC beamline**, a four-plane Delay Wire Chamber (DWC), and a rotatable multi-material PMMA-phantom. The entrance trigger and particle-identification (PID) chain are represented in the Geant4 model as passive upstream material rather than a full detector-response simulation.

Because the delivered secondary beam is mixed, the experiment relies on standard beam tuning, collimation, and the fixed entrance instrumentation: a scintillator coincidence trigger to ensure traversal of the setup (the two paddles serve only as a trigger, not for time-of-flight measurement) together with a gas-threshold Cherenkov veto against lighter relativistic contaminants, defining a proton-enriched sample before DWC reconstruction. The Geant4 model starts from these post-selection protons, and any **common upstream-material contribution is removed differentially with the empty-target calibration**.

Two DWC planes upstream and two downstream provide 60 cm lever arms on each side for reconstruction of the incident and exit tracks. Because the scattering angle is obtained from dot product of those directions, precise beam–phantom perpendicularity is not critical for angle reconstruction.

The simulation scope is fixed by Figure 5: everything downstream of the PID chain is modelled in Geant4 explicitly, while the trigger and Cherenkov stages enter only as passive upstream material whose contribution is removed by the empty-target calibration.

Table 1. *Beam and tracker parameters.*

Parameter	*Value*
Beam momentum	*0.7 GeV/c (T ≈ 232 MeV)*
Momentum spread (simulation)	*1 % (Gaussian)*
Angular divergence (simulation)	*5 mrad*
Spot size (simulation)	*30 mm full width at half maximum (FWHM) (σ = 12.74 mm)*
DWC planes	*4 (z = −100, −40, +40, +100 cm)*
Active area	*10 × 10 cm²*
Modelled material / plane	*200 µm G10 + 50 µm Kapton*
Spatial resolution	*300 µm (Gaussian smearing)*

With the spatial resolution of Table 1, the **combined angular resolution is about 1 mrad, well below the few-mrad material contrast** targeted in this study.

The target is a cylindrical polymethyl-methacrylate (PMMA) body (Ø 10 cm × 5 cm) with four full-thickness inserts (Ø 2 cm) at 3 cm radial offsets (Figure 6), as described in Table 2.



Figure 6. *CAD renderings of the rotatable phantom assembly. (a) Exploded view of the PMMA body with four full-thickness cylindrical inserts: PVC (gold), PEEK (blue), Aluminum (white), and PMMA reference (red). The gear-toothed top plate and central shaft provide 6°-step rotation for tomographic projections. (b) Cruciform alignment and support structure with precision-machined brass locating pins and central bore for the rotation axis.*

Table 2. *Phantom insert properties.*

Insert	*Position*	*Density (g/cm³)*	*X_0 (cm)*	*Role*
PVC	*+x*	*1.40*[†]	*18.20*	*Primary polymer pair (Cl)*
PEEK	*−x*	*1.32*	*31.41*	*Primary polymer pair (C,H,O)*
Aluminum	*+y*	*2.70*	*8.90*	*Strong-scattering reference*
PMMA	*−y*	*1.18*	*34.40*	*Body-matched control*

[†] *The NIST PSTAR compilation lists generic PVC at 1.30 g/cm³. The value adopted here (1.40 g/cm³) corresponds to commercially available unplasticised PVC (uPVC), matching the stock material foreseen for the beam test. The radiation length X_0 is recomputed accordingly.*

These materials are **stopping-power-similar but radiation-length-distinct** (Table 2; chemistry in Section 3, Figure 3).

Four complementary run configurations are used, with the simulated proton histories summarised in Table 3.

Table 3. *Run configurations.*

Mode	*Configuration*	*Proton histories*	*Role*
Full phantom	*All inserts in the PMMA body*	*100,000*	*Primary measurement*
Empty-target calibration	*No phantom*	*50,000*	*Air-subtraction baseline*
Control runs	*PVC-only, PEEK-only, PMMA-body-only*	*100,000 each*	*Single-material validation*
Tomography	*60 projections at 6° increments*	*10,000 per projection*	*Stretch goal: image reconstruction*

The proton counts in Table 3 refer to generated post-selection histories in Geant4; the retained analysis sample is smaller after tracker acceptance and quality cuts.

For each transmitted proton, the simulation records:

- DWC hit coordinates on the four tracking planes;
- reconstructed scattering angle θ and signed distance of closest approach (DCA);
- simulation-side entry and exit kinetic energies, from which WEPL is derived offline;
- material label and phantom rotation angle.

Beamline requirements

We request a proton-enriched secondary beam at 0.7 GeV/c with the standard entrance trigger and particle-identification chain, delivered over an estimated ten days of beam time (full-phantom run, empty-target calibration, single-material control runs, a four-point momentum scan, an insert-reversal systematics run, and a tomographic stretch run; see the plan in Section 6). Table 4 lists what we ask the BBLC facility to provide and what our team brings.

Table 4. *Requested beam and equipment split.*

Provided by the BBLC facility	*Provided by our team*
Proton-enriched 0.7 GeV/c secondary beam (~10 days)	*Rotatable PMMA phantom with PVC, PEEK, Al and PMMA inserts*
Scintillator trigger and threshold Cherenkov PID	*Phantom rotation stage and mounting*
Four-plane DWC telescope and data acquisition	*Geant4 + Python analysis pipeline (already developed)*
Beam tuning, collimation and operations support	*On-site running, calibration and analysis*

5 Simulation Evidence and Expected Results

To show that the proposed measurement will work, we have already carried out a full Geant4 simulation of the experiment; the results in this section are the preparatory evidence that the beam test is designed to reproduce under real conditions. All simulations were performed with Geant4 11.4.0 using FTFP_BERT together with G4EmStandardPhysics_option4. Production cuts were set to 0.1 mm to preserve standard hadronic transport while maintaining the electromagnetic precision required for small-angle scattering [15, 16].

For each transmitted proton, the four DWC hits reconstructed upstream and downstream tracks, from which the 3D scattering angle and signed DCA were obtained. The angular observable is reconstructed as

$$\theta = \arccos(\hat{u}_{\text{in}} \cdot \hat{u}_{\text{out}})$$

where $\hat{u}_{in}$ and $\hat{u}_{out}$ are the normalized upstream and downstream direction vectors. In the full-phantom Mode A run (Table 3), the 100,000 proton histories yielded **82,079 valid four-plane tracks**. The subsequent |DCA| < 3 mm requirement retained 80,988 events, while the empty-target calibration retained 46,899 events. Figure 7 shows the narrow, symmetric DCA core used for this quality cut.

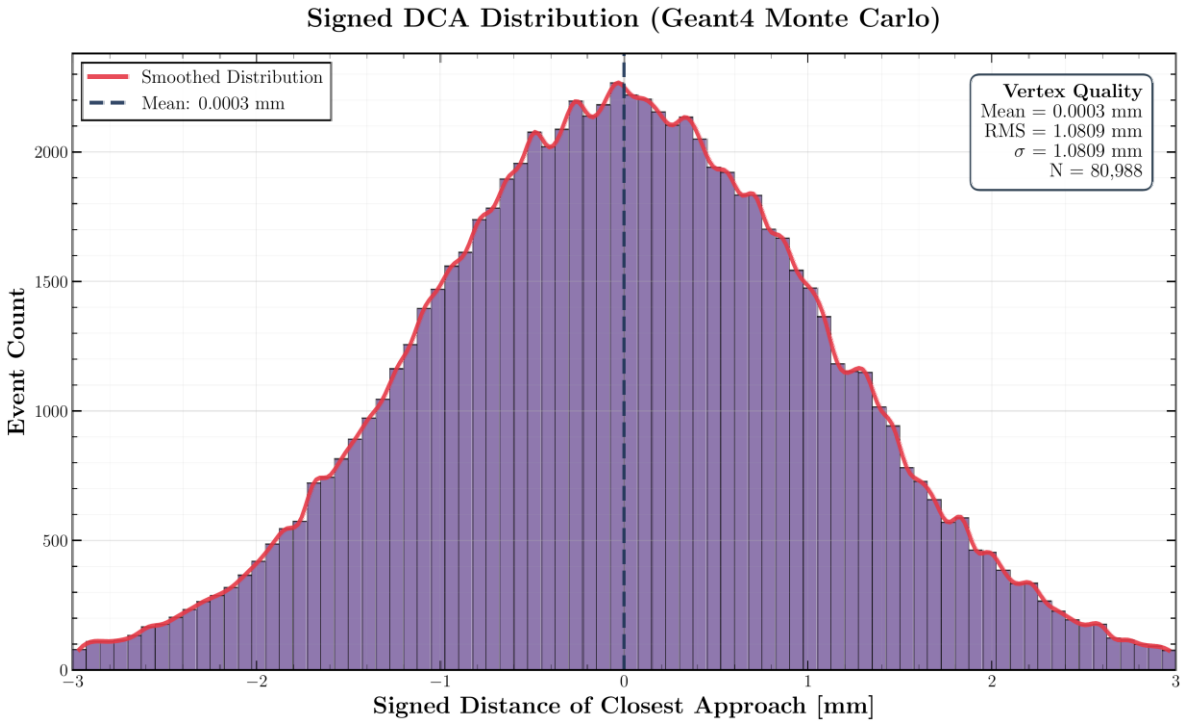


Figure 7. *Signed distance of closest approach (DCA) distribution for reconstructed tracks. The narrow symmetric distribution confirms detector alignment and track reconstruction quality; the |DCA| < 3 mm cut retains the Coulomb-dominated core.*

WEPL was extracted from the Geant4 entry and exit energies by integrating the inverse water stopping power, using the NIST PSTAR water stopping-power data [17]

$$\mathrm{WEPL} = \int_{E_{\mathrm{exit}}}^{E_{\mathrm{entry}}} \frac{dE}{S_{\mathrm{water}}(E)}$$

For the angular width we report the RMS scattering angle after discarding the largest 10% of angles, which removes the rare wide-angle nuclear scatters and leaves the Coulomb-dominated core. Table 5 reports per-material results with this estimator and shows the central numerical finding: PVC and PEEK have **indistinguishable mean WEPL but clearly different scattering widths**, as visible in the smoothed distributions of Figure 8. The WEPL spreads tabulated are single-event σ rather than uncertainties on the mean, so the 0.011 cm gap between PVC and PEEK reflects per-event straggling, not a statistically significant offset between the two materials.

Table 5. *Per-material simulation results. "PMMA body" denotes protons traversing the cylinder body outside any insert; "PMMA reference" denotes the body-matched insert at the –y position. WEPL spreads are per-proton σ, not uncertainties on the mean.*

Material	*Accepted events*	*$\theta_{RMS,tr}$ (mrad)*	*Mean WEPL (cm water)*
PMMA body	*71,389*	*14.60*	*5.92 ± 0.19*
PMMA reference	*2,628*	*14.14*	*5.91 ± 0.18*
PVC	*2,437*	*19.31*	*6.305 ± 0.20*
PEEK	*2,622*	*15.00*	*6.294 ± 0.20*
Aluminum	*1,883*	*26.76*	*10.40 ± 0.99*

The same distinction shows up in the per-material numbers of Table 5: PVC and PEEK overlap in WEPL but separate clearly in scattering width.

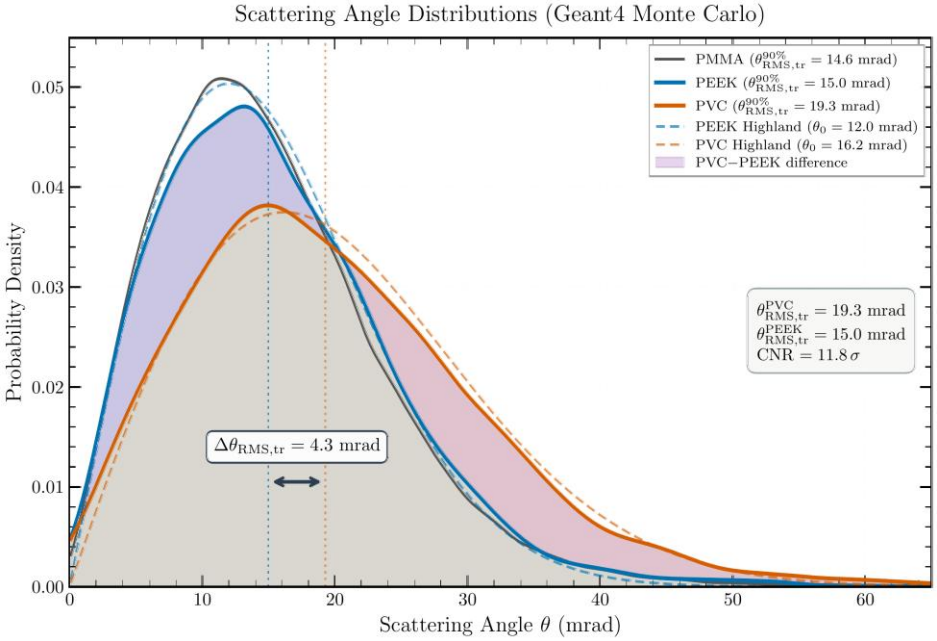


Figure 8. *Smoothed scattering-angle distributions for PVC, PEEK, and PMMA body (Geant4, p = 0.7 GeV/c, t = 5 cm). Truncated RMS values (90th percentile) are annotated for each material.*

The same contrast shows up when we map each channel across the phantom: **the energy-loss map leaves PVC and PEEK merged, whereas the scattering map separates them** (Figure 9).

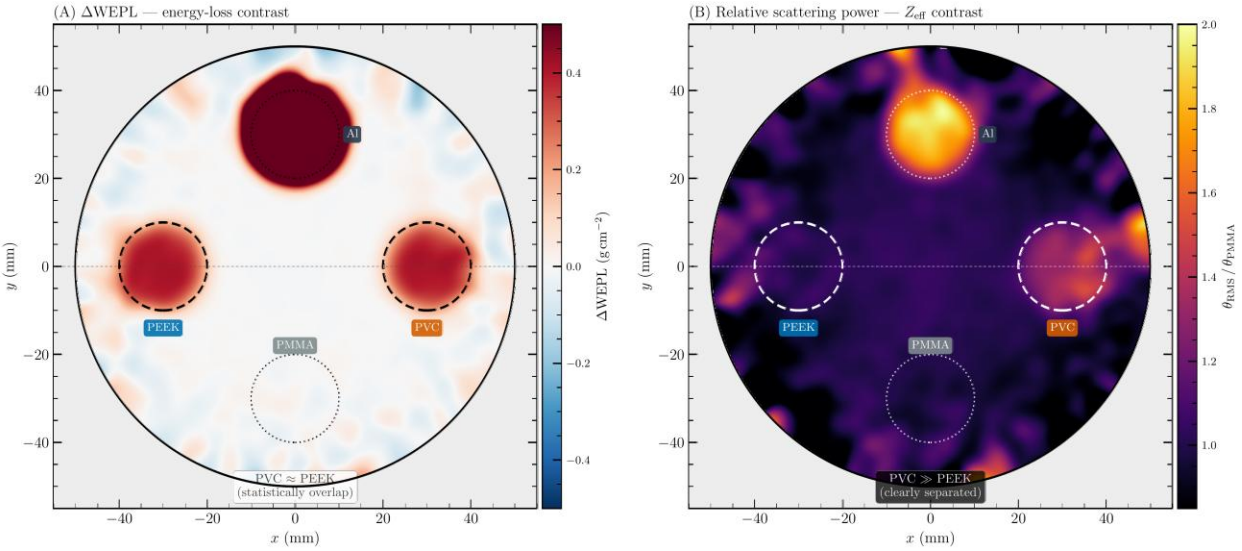


Figure 9. *Two-channel proton transmission map from the full-phantom run (single projection, 0°). (a) Energy-loss (ΔWEPL) map: PVC and PEEK merge. (b) Scattering-width map: PVC and PEEK separate through their radiation-length difference.*

The discrimination metrics in Table 6 are computed from the PVC and PEEK insert populations themselves, not from the PMMA-dominated full field. Each insert retains roughly 2,400 clean tracks, about seven times the statistical floor discussed in Section 6. The measured PVC-to-PEEK width ratio is 1.287, about 29% broader for PVC and only **4.2 % below the 5 cm Highland prediction of 1.343 from Section 3** [8, 13], in good agreement with standard scattering theory. To remove the common upstream and air contribution, the empty-target baseline is removed by subtracting its variance,

$$\Delta\sigma_\theta^2(\mathrm{mat}) = \sigma_{\theta,\mathrm{mat}}^2 - \sigma_{\theta,\mathrm{air}}^2$$

which leaves an air-subtracted PVC/PEEK variance ratio of 1.683. The absolute width difference, 4.31 mrad, is more than four times the 1 mrad angular resolution, so PVC and PEEK separate cleanly, well above the 5σ level that only 339 tracks per material would require (Section 6). Expressed as a variance-based contrast-to-noise ratio, the full-dataset separation is **11.8σ**, the value annotated in Figure 8.

Table 6. *Discrimination results.*

Observable	*Value*	*Interpretation*
WEPL contrast (PVC vs PEEK)	*0.175 %*	*stopping-power channel is nearly degenerate*
Measured $\theta_{RMS,tr}$(PVC)/$\theta_{RMS,tr}$(PEEK)	*1.287*	*PVC broadens the beam more strongly*
Air-subtracted differential variance ratio	*1.683*	*primary scattering contrast metric*
Absolute $\theta_{RMS,tr}$ difference	*4.31 mrad*	*well above 1 mrad angular resolution*
Variance-based CNR (full dataset)	***11.8σ***	*headline statistical separation*

These results confirm the central claim: PVC and PEEK are WEPL-degenerate yet clearly separable in scattering at 0.7 GeV/c, and the main goal becomes demonstrating this scattering contrast under real beamline conditions.

6 Feasibility, Timeline and Risk Assessment

Feasibility is statistics-limited. A **5σ PVC–PEEK separation requires about 339 tracks per material**, whereas the full-phantom 100,000-proton run yields roughly 2,400 clean tracks per insert after the 81% combined acceptance (four-plane tracking and |DCA| < 3 mm) and the ~3% geometric insert fraction (Table 3). That leaves a **safety margin of about seven** over the threshold, and Figure 10 shows that the margin survives realistic per-projection counting noise, where the PVC–PEEK separation still reaches **13σ** at only 2,000 protons per material.

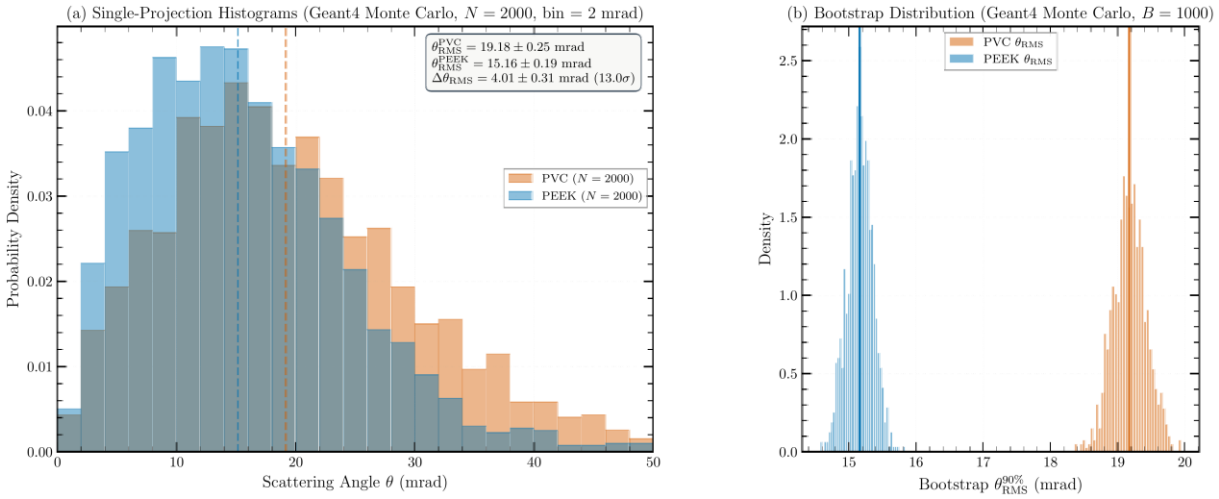


Figure 10. *Finite-statistics Monte Carlo study (N = 2000 protons per material). (a) Raw single-projection histograms showing realistic statistical fluctuations. (b) The same separation appears across many resampled subsets of the simulated tracks, confirming it survives realistic per-projection statistics.*

Two further margins reinforce the statistical case. The **angular margin**: DWC resolution (~1 mrad) is about four times smaller than the 4.3 mrad PVC–PEEK scattering difference (Table 1). The **theoretical margin**: the simulated PVC/PEEK width ratio agrees with Highland to 4.2%, anchoring the scattering model in standard theory. Together with the discrimination metrics in Table 6, the aggregate full-phantom run carries the primary beam-time claim, and the 60-projection tomography scan remains a stretch goal. Table 7 summarises the principal risks and their mitigations.

Table 7. Risk assessment.

Risk	*Mitigation*	*Status*
Mixed-beam contamination (pions, kaons)	*Entrance PID chain + empty-target subtraction*	*Mitigated*
Per-projection tomography statistics (~240 tracks/insert, below 5σ)	*Aggregate full-phantom run carries the primary claim; tomography remains secondary*	*Accepted*
Unmodelled DWC pile-up and inefficiency	*Low beam spill rates; losses act mainly as reduced statistics*	*Accepted*
Phantom machining tolerances	*3 cm offset with 2 cm inserts gives clear separation, well within machining tolerances*	*Mitigated*
Single momentum (0.7 GeV/c only)	*Standard BBLC setting; energy scan is a natural follow-up*	*Accepted*

In aggregate, no risk in Table 7 threatens the main scattering claim; only the secondary tomographic deliverable depends sensitively on margin assumptions about per-projection statistics.

Beam-time plan

Table 8 sets out a proposed ten-day plan. The momentum scan (Day 9) and the insert-reversal systematics run (Day 10) are fixed; the earlier days can be compressed if beam time is shorter, since the primary claim rests only on the full-phantom and calibration runs.

Table 8. *Proposed beam-time plan.*

Day	*Activity*
1	*DWC telescope installation, alignment survey, and DAQ and cabling checkout*
2	*Beam tuning to 0.7 GeV/c; trigger and threshold-Cherenkov PID commissioning; empty-target calibration (50,000 protons)*
3	*Full-phantom production run (100,000 protons) and PMMA-body reference, with online data-quality monitoring*
4	*Single-material control runs (PVC-only, PEEK-only)*
5	*PMMA-only control run, contingency statistics, and first offline analysis pass*
6–7	*Tomographic stretch run (60 projections at 6° steps, 10,000 protons each) with phantom rotation*
8	*Contingency and detector-alignment cross-checks*
9	*Momentum scan: retune to 0.5, 0.8 and 1.0 GeV/c, with phantom and empty-target runs at each*
10	*Systematics: PVC/PEEK insert-reversal run, physics-list cross-checks, and teardown*

Materials and equipment

The team provides a cylindrical PMMA phantom (Ø 10 × 5 cm) with four full-thickness Ø 2 cm inserts (PVC, PEEK, aluminium and a PMMA reference; the unplasticised-PVC stock is noted in Table 2) and a rotation stage for tomographic projections. The BBLC facility provides the beam, the entrance trigger and PID chain, and the four-plane DWC telescope with data acquisition (Table 4). Analysis uses our open-source Geant4

and Python pipeline, which is already developed and validated against the simulation in Section 5.

Radiation safety

The measurement uses a low-intensity secondary beam and introduces no active radioactive sources. All operation follows the facility's standard radiation-protection interlocks, controlled-area access and personal dosimetry under the responsible safety officer; the phantom and inserts are inert plastics and metals that require no special handling.

7 Conclusion and Clinical Relevance

This Geant4 study demonstrates that PVC and PEEK, whose mean WEPL values differ by only 0.175%, can be separated in the scattering channel at 0.7 GeV/c with a **variance-based significance of 11.8σ**, far beyond the 5σ threshold that only 339 tracks per material would require (Table 6). A beam test at the BBLC beamline would confirm that this radiation-length contrast survives under real conditions, validate the Highland-based scattering model against measured angular distributions, and provide the **first detailed scattering measurement of the PVC–PEEK pair at this momentum**.

The broader relevance is proton-range uncertainty in treatment planning. Current clinical margins of 3.5% + 1 mm absorb stopping-power conversion uncertainty [2]. A dual-modality pCT framework reconstructing both stopping and scattering could reduce the composition-dependent component of that budget by resolving material ambiguities that energy-loss imaging alone cannot separate [3, 4]. This is **particularly relevant for paediatric tumours**, where even modest margin reductions spare developing tissue from lifelong treatment toxicities.

8 Sustainable Development Goal Alignment

The project speaks to several United Nations Sustainable Development Goals (SDGs).

SDG 15: Life on Land. The scattering observable studied here is, at heart, a non-destructive material-characterization technique whose physical principle, radiation-length contrast, is not specific to medicine. The same observable underpins cosmic-ray muography of geological and archaeological structures, and is in principle transferable to density and composition mapping of soils, biomass, and contaminated land, which are direct inputs to terrestrial-ecosystem monitoring. Within healthcare itself, more accurate proton-range planning shrinks the volume of irradiated tissue per treatment, lowering the long-term load of activated medical waste returned to the environment.

SDG 3: Good Health and Well-Being. The clinical link of Section 7 is the most direct line to SDG 3: dual-modality pCT can compress the 3.5% + 1 mm proton-range margin, with the largest expected benefit in paediatric oncology, where every spared millimetre of healthy tissue matters across a lifetime.

SDG 4: Quality Education. The work was carried out by secondary-school students under academic mentorship, demonstrating that authentic frontier-physics research (Geant4 simulation, statistical analysis, and beam-line proposal writing) is accessible at the school level when supported by international collaboration.

SDG 9: Industry, Innovation, and Infrastructure. Treating scattering as a primary imaging signal rather than as a noise term is a methodological contribution to the medical-imaging instrument chain; the analysis relies entirely on open-source tools (Geant4 and Python), keeping the method reproducible and transferable to other groups.

SDG 17: Partnerships for the Goals. The international mentor network behind this proposal spans several institutions across multiple continents, made possible by infrastructure (the BBLC beamline, the open-source Geant4 toolkit) explicitly designed for shared use.

9 Creativity of the Experiment

What makes our experiment original is that it deliberately uses what most proton-imaging analyses throw away. Multiple Coulomb scattering is normally treated as a blur to be suppressed; we turn it into the measurement itself. The PVC–PEEK pair is chosen precisely because it defeats the standard energy-loss channel (the two materials are degenerate in WEPL), so any separation *must* come from the new observable, making this a clean, falsifiable test rather than a demonstration of something already known. The approach is also dual-modality: the same protons yield both an energy-loss image and a scattering image, and only the second one tells the materials apart. Finally, the project is genuinely interdisciplinary, combining medical physics, detector instrumentation, Monte-Carlo simulation and statistical data analysis in a single pipeline that we built and validated ourselves.

10 Motivation to Participate in BBLC

We are drawn to this project because it sits right at the meeting point of fundamental physics and medicine: the same scattering that blurs a proton image can, read carefully, become a tool that helps plan safer cancer treatments. Taking part would let us test our own prediction against reality, learn how a real beamline and tracking detectors behave, and experience the full arc of an experiment from idea to data. As Lederman writes in *The God Particle*, the experimentalist's task is to reveal the invisible: proton transport inside matter cannot be seen directly, but by measuring scattering with enough care, we can turn hidden material differences into an observable signal.

11 Team Description

We are a team of four secondary-school students from Oberoi International School, India, working under the guidance of Dr. Chandana Ghosh.

Rizul Garg (Team Leader). Coordinates the project and the beam-test plan, leads the physics case, and integrates the proposal; corresponding author.

Yash Varshney. Geant4 simulation lead, responsible for the detector geometry, physics-list configuration, and run production.

Tranav Tyagi. Data analysis and statistics: track reconstruction, the scattering-width estimators, the discrimination metrics, and the open-source analysis pipeline.

Anshveer Bindra. Phantom and instrumentation design: the CAD of the rotatable phantom and inserts, and the experimental-setup figures.

How we collaborate. We meet weekly to set goals and review progress, share all code and drafts through a common Git repository, and divide the work along each member's strengths while cross-checking one another's results. Physics and analysis decisions are made together, with the team leader coordinating the final write-up.

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